Designation: F 3890-24

Standard Specification for Bioabsorbable Polymer Surgical Mesh

This standard was issued on May 15, 2024; the version of record is continuously maintained.

1. Scope

1.1 This specification covers bioabsorbable polymer mesh for soft tissue reinforcement and repair in surgical procedures

1.2 The bioabsorbable surgical mesh covered by this specification are intended for use in hernia repair, pelvic floor reconstruction, and other soft tissue support applications where temporary reinforcement is needed.

1.3 The values stated in SI units are to be regarded as standard. No other units of measurement are included in this standard.

1.4 This standard does not purport to address all of the safety concerns, if any, associated with its use. It is the responsibility of the user of this standard to establish appropriate safety, health, and environmental practices and determine the applicability of regulatory limitations prior to use.

2. Referenced Documents

2.1 ASTM Standards:

F 2103 Guide for Characterization and Testing of Biomaterials for Surgical Mesh

D 638 Test Method for Tensile Properties of Plastics

F 1635 Test Method for in vitro Degradation Testing of Hydrolytically Degradable Polymer Resins

F 756 Practice for Assessment of Hemolytic Properties of Materials

3. Terminology

3.1 Definitions:

3.1.1 bioabsorbable—capable of being absorbed by living tissue through biological processes

3.1.2 degradation time—the time required for the polymer to lose mechanical integrity and be absorbed by the body

3.1.3 mesh porosity—the percentage of void space in the mesh structure

4. Classification

4.1 bioabsorbable surgical mesh shall be classified according to polymer composition and degradation rate.

4.1.1 **Type I**—Poly(glycolic acid) based, complete absorption in 3-4 months

4.1.2 **Type II**—Poly(lactic acid) based, complete absorption in 12-18 months

4.1.3 **Type III**—Poly(glycolic-co-lactic acid) copolymer, complete absorption in 6-9 months

5. Ordering Information

5.1 Orders for material under this specification shall include the following information:

5.1.1 ASTM designation and year of issue

5.1.2 Type (I, II, or III)

5.1.3 Mesh dimensions (length × width in cm)

5.1.4 Pore size specification

5.1.5 Quantity (number of units)

5.1.6 Sterile or non-sterile

5.1.7 Packaging requirements

6. Materials and Manufacture

6.1 Polymers shall be medical grade with documented biocompatibility testing.

6.2 Materials shall be free from additives, plasticizers, or colorants not approved for implantation.

6.3 Raw materials shall have certificates of analysis demonstrating compliance with specifications.

7. Mechanical and Physical Requirements

Property	Requirement	Test Method
Tensile strength (initial)	≥ 100 N/cm width	ASTM D 638
Elongation at break	30-80%	ASTM D 638
Pore size	0.5-3.0 mm	Microscopy
Porosity	≥ 70%	Calculated

8. Dimensions and Permissible Variations

8.1 Standard mesh dimensions range from 5 cm × 5 cm to 30 cm × 30 cm.

8.2 Dimension tolerance shall be ±5 mm for sides less than 15 cm and ±10 mm for sides greater than 15 cm.

8.3 Thickness shall be 0.3-1.5 mm depending on mesh design.

9. Workmanship, Finish, and Appearance

9.1 Mesh shall be free from tears, holes, and contamination. Edges may be finished or unfinished as specified. Color shall be uniform.

10. Sampling

10.1 Sampling plans shall follow ISO 2859-1 acceptable quality limit procedures. Biocompatibility testing required for each material lot.

11. Test Methods

11.1 **Mechanical Testing**—Test mesh samples in both machine and cross directions per ASTM D 638 adapted for mesh geometry.

11.2 **Degradation Testing**—Evaluate in vitro degradation in phosphate buffered saline at 37°C per ASTM F 1635.

11.3 **Biocompatibility**—Conduct cytotoxicity, sensitization, and irritation testing per ISO 10993 series standards.

12. Inspection

12.1 Inspection of the material shall be made as agreed upon between the purchaser and the seller as part of the purchase contract.

13. Rejection and Rehearing

13.1 Material that fails to conform to the requirements of this specification may be rejected. Rejection should be reported to the producer or supplier promptly and in writing.

14. Certification

14.1 A manufacturer's or supplier's certification shall be furnished to the purchaser that the material was manufactured, sampled, tested, and inspected in accordance with this specification and has met the requirements.

15. Product Marking

15.1 Sterile packages shall be labeled with product name, dimensions, lot number, sterilization method, expiration date, and manufacturer information.

16. Packaging and Package Marking

16.1 Individual meshes shall be packaged in pouches using validated sterilization-compatible materials. Sterilization by ethylene oxide or gamma radiation.

17. Keywords

17.1 bioabsorbable; surgical mesh; hernia; polymer; biodegradable; implant; medical device